Prachi Garella

Contact Information

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Education

University of Houston, College of Natural Sciences and Mathematics, Texas, USA. 2022–Present

- PhD Candidate in the Theoretical Particle Physics Group.
  Advisor: Prof. Claudia Ratti
  Expected completion: June 2027

University of Houston, College of Natural Sciences and Mathematics, Texas, USA. 2022–2025

- Master of Science in Physics.

Durham University, Ustinov College, Durham, UK. 2018–2019

- Postgraduate Diploma in Particles, Strings and Cosmology

Amity University Noida, India. 2015–2018

- B.Sc(Honours) – Physics.

Publications

- **Merging multidimensional equations of state of strongly interacting matter via a statistical mixture**. Phys. Rev. D 113 (2026) 11, 114018. doi:10.1103/pvtc-zdyw. arXiv:2601.07987 [nucl-th].
- **A 4D T'-expanded lattice QCD equation of state with a multi-dimensional 3D-Ising-type critical contribution**. Preprint, arXiv:2610.18830 [nucl-th] (2026).
- **Thermodynamically Consistent Merging of Multidimensional QCD Equations of State**. Conference proceeding for Strangeness in Quark Matter 2026. arXiv:2608.20526 [nucl-th].
- **Studying the QCD Matter produced in Heavy-Ion Collisions using the MUSES Calculation Engine**. arXiv:2606.26326 [nucl-th].

Current Research

1. **PhD Thesis**.
   Advisor: Prof. Claudia Ratti, *University of Houston, USA.*
2. **Holography with Multiple Conserved Charges**.
   Collaborators: Prof. Claudia Ratti, *University of Houston, USA*; Prof. Jorge Noronha, *University of Illinois Urbana-Champaign, USA*; Dr. Mauricio Hippert, *Centro Brasileiro de Pesquisas Fisicas, Brazil.*

Achievements

- **Best Presentation Award** in the High Energy Session at the **University of Houston Research Day 2026**.
- Selected as an **APS Student Ambassador**, 2025–2026.
- **Cullen Graduate Travel Fellowship 2025**.
- **2025 Underrepresented Minority Travel Award**.
- **Graduate Tuition Fellowship 2025**.
- **Best Presentation Award** at the **Houston APS Space City Symposium 2025**.
- **Best Presentation Award** in the High Energy Session at the **University of Houston Research Day 2025**.
- **Public Choice Award** at the **NSM Three Minute Thesis 2024.**

Teaching/ Leadership Experience

- **APS Student Ambassador** 2025–2026
- **Women in Physics Society, APS Chapter President** 2023–2025
- **Lead Teaching Assistant** 2022–2024
- **University Physics II**: Teaching Assistant 2023
- **Undergraduate Classical Mechanics (Physics 1101)**: Teaching Assistant 2022

Talks

- **The Mystery of Time: Is The Universe Stuck Moving Forward?** [Public invited talk]
  After5Society, Patterson Park Patio Bar, Houston, Texas, August 18, 2026.
- **What is the Universe Made of? The Quantum Field Story** [Public invited talk]
  After5Society, The Library, Houston, Texas, April 1, 2026.
- **Merging multidimensional equations of state of strongly interacting matter via a statistical mixture** [Conference talk]
  Strangeness in Quark Matter 2026, Los Angeles, California, March 24, 2026.
- **Merging multidimensional equations of state of strongly interacting matter via a statistical mixture** [Colloquium invited talk]
  Department of Physics, Indian Institute of Technology Delhi, January 15, 2026.
- **QCD Equation of State from Black Hole and Hadronic Physics** [Conference talk]
  APS Division of Nuclear Physics 2025, Chicago, Illinois, October 18, 2025.
- **QCD Equation of State from Holography and HRG** [Conference talk]
  Texas Section of the APS Meeting 2025, University of Houston, Texas, April 2025.
- **QCD Equation of State from Holography and HRG** [Conference talk]
  APS Global Physics Summit 2025, Anaheim, California, March 2025.
- **An Introduction to String Theory Swampland** [Conference talk]
  Texas Section of the APS Meeting 2022, Rice University, Houston, Texas, October 14, 2022.
- **Open problems in Theoretical Physics** [Public invited talk]
  Amity University Noida, India, 2021.
- **On the Foundations of Quantum Mechanics** [Public invited talk]
  Christ University, Bangalore, India, 2020.
- **Gravitational Waves and Blackholes** [Public invited talk]
  Beyond Portals Head Quarters, New Delhi, India, 2018.

Summer/ Winter Schools

- **Nuclear Physics from Multi-Messenger Mergers Summer School**. June 2025
  Indiana University, Indiana, USA.
- **101 Introduction to Cluster Computing**. August–October 2023
  University of Houston's Hewlett Packard Enterprise Data Science Institute (HPE DSI).
- **Qiskit Global Summer School 2023**. July–August 2023
  IBM Quantum.
- **Kavli Asian Winter School on Strings, Particles and Cosmology**. 10–23 January 2022
  International Centre for Theoretical Sciences, TIFR.
- **Dualities in Topology and Algebra**. 01–13 February 2021
  International Centre for Theoretical Sciences, TIFR.
- **Nonperturbative and Numerical Approaches to Quantum Gravity, String Theory and Holography**. 18–22 January 2021
  International Centre for Theoretical Sciences, TIFR.

Science Communication

- The Knowmads Podcast
- Social media science communication with over 100K followers on Instagram